

TWENTE, Netherlands

WMO: 062900

Lat: 52.2731N

Lon: 6.8908E

Elev: 35

StdP: 100.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-6.0	-11.8	1.4	-6.6	-9.1	1.7	-4.9	11.6	8.6	10.3	8.3	2.3	60	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.6	20.0	27.4	19.3	25.3	18.3	21.2	27.5	20.2	25.7	19.2	24.1	3.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.5	18.1	13.1	22.2	17.3	12.4	21.2	61.5	27.5	57.9	25.7	54.5	24.2	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.1	7.2	DB	-11.7	33.5	3.7	2.0	-14.4	34.9	-16.6	36.1	-18.7	37.2	-21.4	38.7	(4)
(5)			WB	-12.1	23.4	3.6	0.9	-14.7	24.0	-16.8	24.6	-18.9	25.0	-21.5	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	2.6	3.2	5.6	9.3	13.0	15.7	18.0	17.6	14.3	10.6	6.4	3.4	(6)
(7)		DBStd	6.64	4.47	4.16	3.70	3.97	3.84	3.36	3.30	3.07	3.02	3.61	3.61	4.43	(7)
(8)		HDD10.0	1001	231	192	141	59	14	1	0	0	2	37	117	207	(8)
(9)		HDD18.3	3127	489	424	393	270	171	93	46	52	126	241	358	464	(9)
(10)		CDD10.0	1006	1	1	6	39	106	172	249	235	131	56	8	2	(10)
(11)		CDD18.3	92	0	0	0	1	5	14	37	28	6	1	0	0	(11)
(12)		CDH23.3	939	0	0	0	13	62	165	387	267	44	2	0	0	(12)
(13)		CDH26.7	268	0	0	0	1	10	41	127	83	7	0	0	0	(13)
(14)	Wind	WSAvg	3.4	4.0	3.9	3.8	3.3	3.2	3.1	2.9	2.8	2.9	3.3	3.4	3.9	(14)
(15)	Precipitation	PrecAvg	844	74	62	57	47	62	70	88	86	74	70	75	80	(15)
(16)		PrecMax	1097	132	132	126	105	134	143	196	219	148	217	143	179	(16)
(17)		PrecMin	651	6	22	11	1	24	9	27	19	7	20	8	26	(17)
(18)		PrecStd	114	34	32	34	22	27	34	37	47	36	36	32	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.8	18.8	25.2	28.2	31.0	33.1	32.1	27.7	22.3	16.4	13.3	(19)
(20)			MCWB	10.6	9.3	11.9	14.8	18.4	20.9	20.6	20.9	19.4	17.2	13.6	11.4	(20)
(21)		2%	DB	11.1	11.5	15.7	21.8	24.9	27.3	30.0	29.0	24.3	19.3	14.2	12.0	(21)
(22)			MCWB	9.6	9.0	10.5	13.7	16.1	19.0	20.2	19.9	18.4	15.6	12.4	10.6	(22)
(23)		5%	DB	9.7	10.3	13.1	19.2	22.8	25.0	27.5	26.3	21.6	17.2	12.9	10.7	(23)
(24)			MCWB	8.4	8.5	9.5	12.5	15.4	18.2	19.7	19.2	16.9	14.6	11.4	9.4	(24)
(25)		10%	DB	8.3	9.0	11.4	16.6	20.6	22.7	25.1	23.8	19.5	15.8	11.6	9.3	(25)
(26)			MCWB	7.1	7.4	8.5	11.5	14.7	16.9	18.6	18.0	15.7	13.7	10.1	8.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.8	12.9	16.0	19.5	21.9	22.7	22.2	20.3	17.6	14.0	12.0	(27)
(28)			MCD B	12.0	12.3	16.7	23.1	25.4	28.0	29.1	29.0	25.5	21.3	14.9	12.8	(28)
(29)		2%	WB	9.8	9.6	11.4	14.3	17.8	20.2	21.4	20.9	18.7	16.0	12.7	10.8	(29)
(30)			MCD B	10.9	10.9	14.3	19.8	23.1	25.7	27.8	26.9	23.2	18.6	14.0	11.9	(30)
(31)		5%	WB	8.4	8.6	10.1	13.2	16.4	18.9	20.3	19.7	17.6	15.0	11.5	9.6	(31)
(32)			MCD B	9.5	10.1	12.5	17.9	21.0	23.8	26.1	24.9	20.9	16.8	12.8	10.6	(32)
(33)		10%	WB	7.2	7.5	8.9	12.0	15.1	17.6	19.2	18.7	16.5	14.0	10.2	8.2	(33)
(34)			MCD B	8.2	8.9	11.1	16.1	19.4	21.6	24.0	23.0	19.0	15.6	11.4	9.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.9	7.9	10.4	10.4	10.5	10.4	10.2	9.1	7.6	5.5	4.6	(35)
(36)			MCD BR	5.1	7.2	11.4	14.1	13.7	14.1	14.5	13.9	11.9	8.9	6.2	4.8	(36)
(37)			MCWBR	4.6	5.3	7.3	7.4	6.8	7.2	6.5	6.4	6.6	5.8	4.9	4.2	(37)
(38)		5% WB	MCD BR	5.0	6.2	9.7	12.4	12.2	12.4	12.9	12.5	10.6	8.4	5.9	4.9	(38)
(39)			MCWBR	4.8	5.0	6.7	7.0	6.7	7.0	6.2	6.4	6.2	5.8	5.0	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.325	0.347	0.376	0.408	0.411	0.409	0.420	0.413	0.389	0.361	0.340	0.321	(40)	
(41)		taud	2.356	2.341	2.271	2.222	2.250	2.272	2.249	2.290	2.349	2.405	2.371	2.340	(41)	
(42)		Ebn at Noon	664	759	806	824	841	845	826	811	786	734	637	597	(42)	
(43)		Edh at Noon	62	82	107	126	129	127	128	117	98	77	61	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.36	2.48	4.02	4.90	5.21	4.99	4.19	3.05	1.78	0.82	0.53	(44)	
(45)		RadStd	0.08	0.24	0.34	0.43	0.44	0.41	0.63	0.36	0.35	0.17	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.32	N/A	+1.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page